

KAIHUAI DENG

No.219, Ningliu Road, Nanjing, Jiangsu, China(Postcode: 210044)

(+86) 173-2700-6642 · <https://othodeng.github.io> · kaihuaideng@nuist.edu.cn

EDUCATION

Nanjing University of Information Science and Technology *Sep 2023 – Jul 2027 (expected)*

- **B.S.** in Atmospheric Science, Hydrometeorology division.
- **Weighted average score:** 85.99 | **GPA:** 3.618.

AREA OF INTEREST

Extreme events · Flash droughts · Land–atmosphere coupling · WRF modeling

PUBLICATIONS

- [1] **Deng, K.**, Huang, Z.*, Li, H., Wu, J., (2026). **Global unequal exposure to intensifying hydrological droughts.** *Journal of hydrology*. (Major Revision)
- [2] **Deng, K.**, Huang, Z.*, (2026). **Intensifying global 3D terrestrial water storage drought under climate change.** (To be submit)
- [3] **Deng, K.*** and Wang, J. (2024). **A Model that Explains the Contrasting SST Trends in the Southern Pacific Ocean.** *Science and Technology of Engineering, Chemistry, and Environmental Protection*. doi.org/10.61173/xnag2k18

RESEARCH EXPERIENCE

Global Spatiotemporal Tracking of 3D Hydrological Drought & Exposure *2025.09 – Present*

Independent research. Supervised by Prof. Zhongwei Huang (NUIST) Nanjing

- Processed 22 years of GRACE/GRACE-FO satellite data via EEMD decomposition to calculate **Drought Severity Index(DSI)**.
- Engineered a 3D spatiotemporal **Python clustering algorithm** to track drought life-cycles, revealing anthropogenic climate change drives **64.5%** of worsening water-scarcity exposure.

WRF-based Precipitation Sensitivity Experiments (Jing–Jin–Ji Region) *2025.03 – 2025.07*

Independent research. Supervised by Dr. Mengya Wang (NUIST) Nanjing

- Executed precipitation modeling using **WRF/WPS** on Linux HPC clusters, modifying **cumulus and microphysics parameterizations** to evaluate surface boundary layer fluxes.
- Quantified model uncertainties to identify **optimal parameterization schemes**, minimizing biases in simulated precipitation and surface-atmosphere energy exchanges.

Precipitation–Runoff Prediction in the Yiluo River Basin *2024.06 – 2025.03*

Research assistant. Supervised by Prof. Xin Yuan (LabESM, CAS) Nanjing

- Analyzed historical hydrometeorological forcings (ERA5, CN05.1) utilizing **EOF decomposition** to extract dominant variability modes.

- Developed a **physical-statistical surface water routing model** to translate atmospheric boundary conditions into quantitative streamflow, constructing a robust **basin-scale runoff prediction framework**.

A Model Explaining Contrasting SST Trends in the Southern Pacific 2024.03 – 2025.09

Supervised by Prof. Jun Yin (NUIST) & Prof. George Tselioudis (Columbia University) Beijing

- Analyzed 64 years of **ERA5 reanalysis data** (1960–2024) via **nonequilibrium fluctuation** and **Linear response theory** to evaluate global temperature anomalies.
- Processed multi-decadal **SST data** to quantify Southern Pacific trends, proposing a **mechanism-based model** elucidating regional thermal behaviors based on **coupled sea–air–ice processes**.

INTERNSHIP

University of Prince Edward Island, Mitacs 2026.07 ~ 2026.10 (Expected)

Research assistant. Supervised by Prof. Xander Wang (Climate Smart Lab, UPEI) PEI, Canada

- Conduct comprehensive **Climate Change and Health Vulnerability Assessment** for PEI across risk areas using baseline health data and socioeconomic determinants.
- Map climate-related health disparities for high-risk demographics to inform local **adaptation planning** and public health policy formulation.

Leizhou Meteorological Bureau 2025.07 ~ 2025.08

Intern, Forecasting Division, Guangdong Provincial Meteorological Observation Zhanjiang

- Executed real-time radar/satellite monitoring of severe weather (**Typhoons**) and conducted data QA/QC for regional station networks to support **disaster forecasting**.
- Calibrated and maintained field hydrologic equipment (groundwater wells, flux towers, snow sensors) utilizing **telemetry systems**, ensuring continuous high-fidelity environmental data collection.

SELECTED AWARDS & HONORS

Chinese Government Scholarship. *Selected for Mitacs research internship, CSC.* Jan 2026

First-Class Academic Scholarship. *Recognition for academic performance, NUIST* Nov 2025

Second-Class Academic Scholarship. *Recognition for academic performance, NUIST* Nov 2024

Merit Student. *Recognition for all-round excellence, NUIST* Dec 2024

ADDITIONAL INFORMATION

- Programming: **Python**, Shell, Fortran, LaTeX, Typst.
- Software: Linux (HPC), Git, **WRF**, Excel, Matlab, **Google Earth Engine**.
- Languages: English (IELTS overall 7.0), Chinese (native), French (A2, Duolingo).